From Frequency Response to Bode

EGR 345

GVSU

Outline

- quiz solution
- review
- from frequency response to Bode plots
  - from C_1 and C_2 to M and ϕ
  - what if we do lots of frequencies? (LA 3)
  - are we ready for `semilogx` and dB magnitude?

Review 1

- if we give a sine wave input to a stable system, what happens?

Sine wave in → Sine wave out @ steady-state

Review 1B

- $u(t) = \sin(\omega t)$
- $G(s)$ is stable
- $y_{ss}(t) = ?$

$C_1 \sin(\omega t) + C_2 \cos(\omega t) = M \sin(\omega t + \phi)$

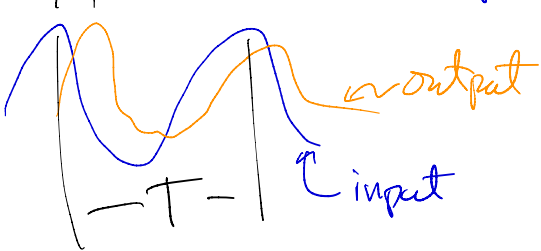
Review 2

- if the input frequency changes, how does y_{ss} change?
  - what did you observe in lab?
    - simulation?
    - experiment?

Magnitude depends on ω
- large near resonance

phase also changes with ω somehow

$\phi° = \frac{-t}{T} \cdot 360°$



OTT Magnitude and phase depend on input frequency

⇒ exactly how they change with input frequency depends on the TF

Review 3

- how did we find C_1 and C_2?

$$G(s) = \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$u(t) = \sin(\omega t)$$

$$U(s) = \frac{\omega}{s^2 + \omega^2}$$

$$Y(s) = \left(\frac{\omega}{s^2+\omega^2} \cdot \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2}\right) = \frac{C_1\omega + C_2 s}{s^2+\omega^2} + \frac{C_3\omega_d + C_4(s+a)}{(s+a)^2 + \omega_d^2}\Bigg](s^2+\omega^2)\Big|_{s=j\omega}$$

$$Y(s) = \frac{\omega}{s^2+\omega^2} \cdot \frac{1/\mu}{s^2+2\beta\omega_n s+\omega_n^2} = \frac{C_1\omega + C_2 s}{s^2+\omega^2} + (\text{other stuff})$$

G

$$G(j\omega)\cdot\omega = C_1\omega + C_2 j\omega$$

$$C_1 = \text{Real}(G(j\omega))$$

$$C_2 = \text{Imag}(G(j\omega))$$

How do we find magnitude and phase?

- M is magnitude and ϕ is phase

$$y_{ss}(t) = C_1 \sin(\omega t) + C_2 \cos(\omega t) = M \sin(\omega t + \phi)$$

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

$$M\sin(\omega t+\phi) = M\sin(\omega t)\cos\phi + M\cos(\omega t)\sin\phi$$

$$C_1 = M\cos\phi \qquad C_2 = M\sin\phi$$

$C_1 = M\cos\phi \qquad C_2 = M\sin\phi$

$$C_1^2 + C_2^2 = M^2(\cos^2\phi + \sin^2\phi) \rightarrow M = \sqrt{C_1^2 + C_2^2}$$

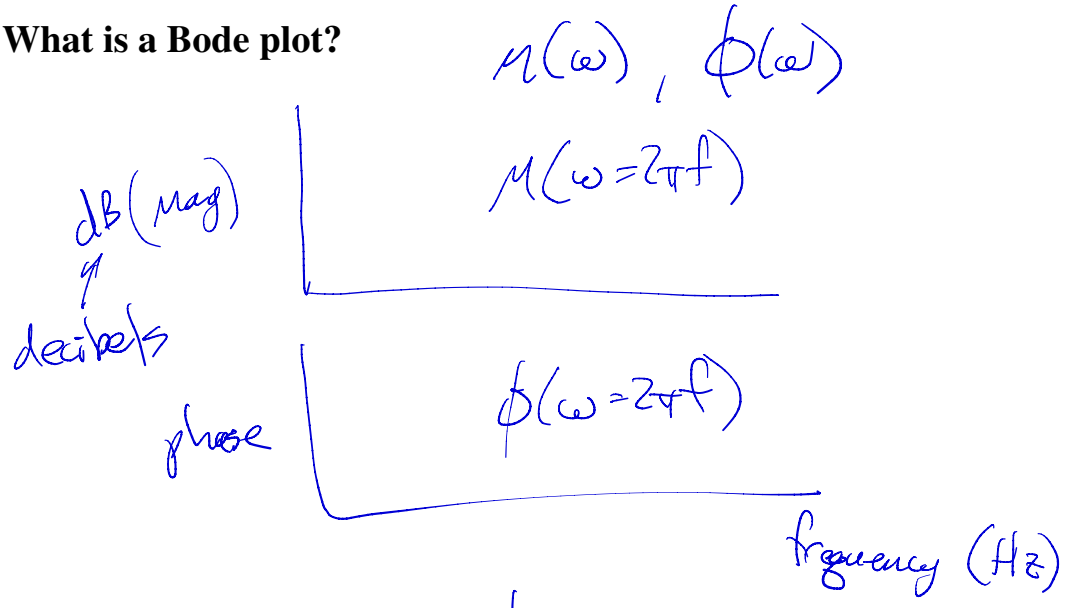
$$\frac{C_2}{C_1} = \frac{M\sin\phi}{M\cos\phi} = \tan\phi$$

$$M = |G(j\omega)|$$

$$\phi = \angle G(j\omega)$$

$G(j\omega) = Me^{j\phi}$ ← polar coordinates

What is a Bode plot?



LA 3

- let’s do this